MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY CHAPTER TEST – 01 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If one of the zeroes of the quadratic polynomial $(k-1)x^2 + kx + 1$ is -3, then the value of k is: [1]

(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Let $p(x) = (k-1)x^2 + kx + 1$, where $k - 1 \neq 0$. We know that if $x = \alpha$ is a zero of $p(x)$, then:

Formula Used: $p(\alpha) = 0$

Since $x = -3$ is a zero of $p(x)$, we substitute $x = -3$:

$$\begin{aligned} p(-3) = 0 &\implies (k-1)(-3)^2 + k(-3) + 1 = 0 \qquad \text{[0.5M]} \\ &\implies 9(k-1) - 3k + 1 = 0 \\ &\implies 9k - 9 - 3k + 1 = 0 \\ &\implies 6k - 8 = 0 \implies k = \frac{8}{6} = \boxed{\frac{4}{3}}. \qquad \text{[0.5M]} \end{aligned}$$

Q2. A quadratic polynomial, the zeroes of which are -3 and 4, is: [1]

(A) $x^2 - x + 12$ (B) $x^2 + x + 12$ (C) $\frac{x^2}{2} - \frac{x}{2} - 6$ (D) $2x^2 + 2x - 24$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Let $\alpha = -3$ and $\beta = 4$. We know that a quadratic polynomial with zeroes α and β is given by:

Formula Used: $p(x) = k\left[x^2 - (\alpha+\beta)x + \alpha\beta\right], \quad k \neq 0$

Computing the sum and product of zeroes:

$$S = \alpha + \beta = -3 + 4 = 1, \qquad P = \alpha\beta = (-3)(4) = -12. \qquad \text{[0.5M]}$$

Hence $p(x) = k(x^2 - x - 12)$. Setting $k = \frac{1}{2}$ yields $\boxed{\frac{x^2}{2} - \frac{x}{2} - 6}$. [0.5M]

Q3. If the zeroes of the quadratic polynomial $x^2 + (a+1)x + b$ are 2 and -3, then: [1]

(A) $a = -7,\ b = -1$ **(B)** $a = 5,\ b = -1$

(C) $a = 2,\ b = -6$ **(D)** $a = 0,\ b = -6$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Comparing $p(x) = x^2 + (a+1)x + b$ with $Ax^2 + Bx + C$, we have $A = 1$, $B = a+1$, and $C = b$. We know Vieta's relations for zeroes $\alpha = 2$ and $\beta = -3$:

Formula Used: $\alpha + \beta = -\dfrac{B}{A}, \qquad \alpha\beta = \dfrac{C}{A}$

Applying both relations systematically:

$$\alpha + \beta = 2 + (-3) = -\frac{a+1}{1} \implies -1 = -(a+1) \implies a + 1 = 1 \implies \boxed{a = 0}, \quad \textbf{[0.5M]}$$

$$\alpha\beta = (2)(-3) = \frac{b}{1} \implies \boxed{b = -6}. \quad \textbf{[0.5M]}$$

Q4. The total number of polynomials having zeroes as -2 and 5 is: [1]

(A) 1 (B) 2 (C) 3 (D) more than 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

We know that any polynomial with zeroes $\alpha = -2$ and $\beta = 5$ can be written as a nonzero constant multiple of $(x - \alpha)(x - \beta)$:

Formula Used: $p(x) = k(x - \alpha)(x - \beta), \quad k \in \mathbb{R} \setminus \{0\}$

Substituting $\alpha = -2$ and $\beta = 5$:

$$p(x) = k(x+2)(x-5) = k(x^2 - 3x - 10). \qquad \textbf{[0.5M]}$$

Since k can take infinitely many nonzero real values (even without higher-degree multiples), there are infinitely many such polynomials, i.e., $\boxed{\text{more than 3}}$. **[0.5M]**

Q5. Assertion-Reason: [1]

Assertion (A): The graph of $p(x) = x^2 - 4x + 4$ touches the x-axis at exactly one point $(2, 0)$.

Reason (R): A quadratic polynomial has two equal zeroes if and only if $D = b^2 - 4ac = 0$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know that for $p(x) = ax^2 + bx + c$, real zeroes correspond to the x-intercepts and repeated zeroes correspond to tangency:

Formula Used: $D = b^2 - 4ac = 0 \iff \alpha = \beta = -\dfrac{b}{2a}$

For $p(x) = x^2 - 4x + 4$, $a = 1$, $b = -4$, $c = 4$. Thus:

$$D = (-4)^2 - 4(1)(4) = 16 - 16 = 0, \quad \text{and} \quad p(x) = (x-2)^2. \qquad \textbf{[0.5M]}$$

Hence $p(x)$ has two equal zeroes at $x = 2$, so the upward parabola touches the x-axis only at $(2, 0)$. Both **(A)** and **(R)** are true, and **(R)** is the correct explanation of **(A)**. **[0.5M]**

Q6. Find the zeroes of the quadratic polynomial $f(u) = 4u^2 + 8u$ and verify the relationship between the zeroes and the coefficients. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $f(u) = 4u^2 + 8u + 0$ with $au^2 + bu + c$, we have $a = 4$, $b = 8$, and $c = 0$. We know that the zeroes satisfy $f(u) = 0$ and Vieta's relations:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Factorisation: Setting $f(u) = 0$:

$$4u^2 + 8u = 0 \implies 4u(u+2) = 0 \implies u = 0 \text{ or } u = -2.$$

Thus the zeroes are $\alpha = \boxed{0}$ and $\beta = \boxed{-2}$. **[1M]**

Step 2 – Verification:

$$\text{Sum of zeroes} = \alpha + \beta = 0 + (-2) = -2, \qquad -\frac{b}{a} = -\frac{8}{4} = -2 \implies \alpha + \beta = -\frac{b}{a}, \quad \textbf{[0.5M]}$$

$$\text{Product of zeroes} = \alpha\beta = (0)(-2) = 0, \qquad \frac{c}{a} = \frac{0}{4} = 0 \implies \alpha\beta = \frac{c}{a}. \quad \textbf{[0.5M]}$$

Hence the relationship between the zeroes and coefficients is verified.

Q7. Find the zeroes of the quadratic polynomial $f(x) = 6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Rewriting $f(x)$ in standard descending powers of x: $f(x) = 6x^2 - 7x - 3$. Comparing with $ax^2 + bx + c$, we get $a = 6$, $b = -7$, and $c = -3$. We know the standard coefficient relations:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Splitting the middle term: Since $a \cdot c = 6(-3) = -18$ and $(-9) + 2 = -7$, $(-9)(2) = -18$:

$$f(x) = 6x^2 - 9x + 2x - 3 = 3x(2x-3) + 1(2x-3) = (3x+1)(2x-3). \qquad \textbf{[1M]}$$

Setting $f(x) = 0$ gives $3x + 1 = 0$ or $2x - 3 = 0$, so the zeroes are $\alpha = \boxed{-\dfrac{1}{3}}$ and $\beta = \boxed{\dfrac{3}{2}}$. **[1M]**

Step 2 – Verification:

$$\text{Sum} = \alpha + \beta = -\frac{1}{3} + \frac{3}{2} = \frac{-2+9}{6} = \frac{7}{6}, \qquad -\frac{b}{a} = -\frac{(-7)}{6} = \frac{7}{6}, \qquad \textbf{[0.5M]}$$

$$\text{Product} = \alpha\beta = \left(-\frac{1}{3}\right)\left(\frac{3}{2}\right) = -\frac{1}{2}, \qquad \frac{c}{a} = \frac{-3}{6} = -\frac{1}{2}. \qquad \textbf{[0.5M]}$$

Since $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$, the relations are verified.

Q8. Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $\frac{1}{3}$ respectively. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the zeroes of the required quadratic polynomial be α and β. Given:

$$S = \alpha + \beta = \sqrt{2}, \qquad P = \alpha\beta = \frac{1}{3}. \qquad \text{[1M]}$$

We know that a quadratic polynomial with given sum S and product P of zeroes is:

Formula Used: $p(x) = k\big(x^2 - Sx + P\big), \quad k \neq 0$

Substituting $S = \sqrt{2}$ and $P = \frac{1}{3}$:

$$p(x) = k\left(x^2 - \sqrt{2}\,x + \frac{1}{3}\right) \qquad \text{[1M]}$$
$$= \frac{k}{3}\left(3x^2 - 3\sqrt{2}\,x + 1\right).$$

Choosing $k = 3$ (to eliminate fractional coefficients), a clean required quadratic polynomial is:

$$\boxed{p(x) = 3x^2 - 3\sqrt{2}\,x + 1} \quad \text{(or any nonzero multiple } c(3x^2 - 3\sqrt{2}\,x + 1)\text{).} \qquad \text{[1M]}$$

Q9. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - 5x + 4$, find the value of: $\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $f(x) = x^2 - 5x + 4$ with $ax^2 + bx + c$, we have $a = 1$, $b = -5$, and $c = 4$. We know Vieta's sum and product identities:

Formula Used: $\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

Thus:

$$\alpha + \beta = -\frac{(-5)}{1} = 5, \qquad \alpha\beta = \frac{4}{1} = 4. \qquad \text{[1M]}$$

Taking the LCM for the fractional part of the required expression:

$$\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta = \frac{\beta + \alpha}{\alpha\beta} - 2\alpha\beta \qquad \text{[1M]}$$
$$= \frac{5}{4} - 2(4) = \frac{5}{4} - 8 = \frac{5 - 32}{4} = \boxed{-\frac{27}{4}}. \qquad \text{[1M]}$$

Q10. Case Study: Parabolic Highway Underpass

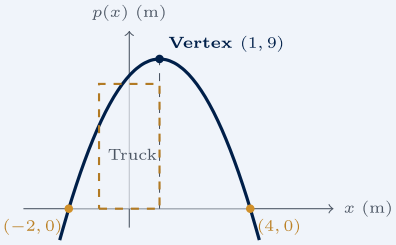
A highway underpass has a parabolic arch profile represented by $p(x) = -x^2 + 2x + 8$, where x is horizontal distance (in m) from the centre line and $p(x)$ is clearance height (in m) above the road level. [4]

(i) Find the zeroes of $p(x) = -x^2 + 2x + 8$. What do these zeroes represent physically? **[1]**

(ii) What is the maximum clearance height of the underpass, and at what horizontal position does it occur? **[1]**

(iii) **(a)** If a second underpass nearby has zeroes at -3 m and 5 m, find the quadratic polynomial representing its shape. **OR (b)** A cargo truck of width 2 m ($x = -1$ to $x = 1$) and height 7.5 m enters along the central lane. Will it clear safely? **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given Model & Arch Geometry: The clearance function $p(x) = -x^2 + 2x + 8$ has $a = -1 < 0$, forming a downward-opening parabola centered at vertex $(1, 9)$. Zeroes lie asymmetrically at $x = -2$ m and $x = 4$ m, so clearance along $x \in [-1, 1]$ attains its lowest value at the left edge $x = -1$.

Formula Used: $p(x) = -x^2 + 2x + 8 = -(x-1)^2 + 9, \qquad q(x) = k(x^2 - Sx + P)$

(i) Zeroes & physical meaning: Setting $p(x) = 0$:

$$-x^2 + 2x + 8 = 0 \implies x^2 - 2x - 8 = 0 \implies (x-4)(x+2) = 0 \implies x = \boxed{-2 \text{ m}}, \ \boxed{4 \text{ m}}.$$

Physically, $x = -2$ m and $x = 4$ m locate the two base edges where the underpass arch meets the road level (total base width $4 - (-2) = 6$ m). [1M]

(ii) Maximum clearance height: Completing the square in Class 10 form:

$$p(x) = -(x^2 - 2x) + 8 = -[(x-1)^2 - 1] + 8 = -(x-1)^2 + 9.$$

Since $(x-1)^2 \geq 0$, maximum clearance is $\boxed{9 \text{ m}}$ at $x = \boxed{1 \text{ m}}$ right of the centre line. [1M]

(iii)(a) Second underpass polynomial: Zeroes are $\alpha = -3, \beta = 5 \implies S = 2, P = -15$, so $q(x) = k(x^2 - 2x - 15)$. Taking $k = -1$ for a downward arch gives $\boxed{-x^2 + 2x + 15}$. [2M]

OR (Alternative Part (b) – Truck clearance check): Over $x \in [-1, 1]$, minimum clearance occurs at the left edge $x = -1$:

$$p(-1) = -(-1)^2 + 2(-1) + 8 = -1 - 2 + 8 = \boxed{5 \text{ m}}. \qquad [1M]$$

(At $x = 0$, $p(0) = 8$ m; at $x = 1$, $p(1) = 9$ m.) Since 5 m < 7.5 m, the top-left corner strikes the arch (short by 2.5 m), so $\boxed{\text{No, the truck will NOT clear safely.}}$ [1M]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★